The empirical recurrence for OEIS A251032, recovered from its tail

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Abstract

OEIS A251032 records "Empirical recurrence of order 59" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 217$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 61$. Its last coefficient is $c_{59} = 6945523200 \neq 0$, so no further tail satisfies anything shorter; any order-59 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A251032 is "Number of (n+1)X(3+1) 0..3 arrays with nondecreasing min(x(i,j),x(i,j-1)) in the i direction and nondecreasing absolute value of x(i,j)-x(i-1,j) in the j direction.". It has offset 1 and begins

$$4050, 45560, 422538, 3396593, 25346175, 179902335, 1238480366, 8354406377, \ \ldots$$

The entry states:

> Empirical recurrence of order 59 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 13:01:11, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 217$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 217$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^{\top} M^{j}$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. □

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 59: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 59.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 434$ exact terms from $n = 2$ on, it returns order 59 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{59} c_i\, a(n-i),$$

c_1	75	c_{16}	−15804987481962	c_{31}	175415746924320162	c_{46}	32978669662746391
c_2	−2629	c_{17}	−30906134963268	c_{32}	−85184623861560077	c_{47}	4993140022930373
c_3	56993	c_{18}	174094469346612	c_{33}	−215481757438592709	c_{48}	−10269991107252896
c_4	−850697	c_{19}	−224766016159022	c_{34}	210430389452348067	c_{49}	−3118991368550570
c_5	9184445	c_{20}	−400768871305922	c_{35}	184920799596413637	c_{50}	2063608959431486
c_6	−72902639	c_{21}	1749614925696370	c_{36}	−297157484272840349	c_{51}	1130587893799432
c_7	418104929	c_{22}	−1319904550926190	c_{37}	−108545816719805231	c_{52}	−137411325306640
c_8	−1583734876	c_{23}	−4421423239620742	c_{38}	302473553806080677	c_{53}	−220958703324904
c_9	2384031212	c_{24}	10511850378345488	c_{39}	38739259959025653	c_{54}	−41186812558328
c_{10}	13692331091	c_{25}	335774574379552	c_{40}	−239602425026403228	c_{55}	13134910895856
c_{11}	−112396993759	c_{26}	−29534715693057274	c_{41}	−4914632411792356	c_{56}	7815424247040
c_{12}	370106392076	c_{27}	29174284309591442	c_{42}	152995272556984919	c_{57}	1634772485952
c_{13}	−334417580020	c_{28}	41567148695318632	c_{43}	25098758290189	c_{58}	166742547840
c_{14}	−2347040694765	c_{29}	−95700487120627704	c_{44}	−79503245237930748	c_{59}	6945523200
c_{15}	10953875298777	c_{30}	−10033043501311098	c_{45}	−3849854312017964		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 61$, and it is the recurrence OEIS A251032 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{59} - c_1 x^{59-1} - \cdots - c_{59}$. Its constant term is $-c_{59} = -6945523200$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 59 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 59, so $\deg q = 59$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 59, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 17 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 59, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 6945523200.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-59 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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